LEARNER GUIDE

Tangents to a circle

Circle Theorems · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A **tangent** touches a circle at exactly one point, the **point of contact**. Two clean facts govern tangents. First, a tangent is **perpendicular to the radius** drawn to the point of contact — it meets it at **90°** — because that radius is the shortest line from the centre O to the tangent, and the shortest line from a point to a line is always perpendicular. Second, the **two tangents from an external point P are equal in length**. The figure OAPB (two radii, two tangents) is a **kite**, symmetrical about OP, and the two right-angled triangles OAP and OBP are **congruent** — so PA = PB. A useful consequence: in that kite, ∠AOB + ∠APB = 180°.

What you are learning to notice

- A **tangent meets the radius at 90°** at the point of contact.
- The **two tangents from an external point are equal** (PA = PB).
- The figure OAPB is a **kite**; ∠AOB + ∠APB = 180°.
- Both facts come from equal radii and congruent right-angled triangles.

Moving through it, one step at a time

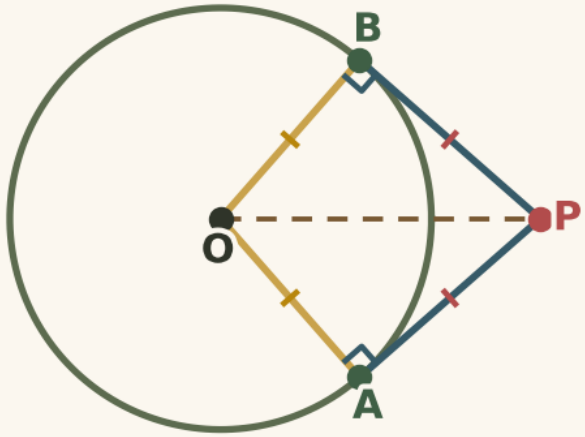
Look for a radius drawn to a point of contact — that is a hidden right angle. Look for two tangents from one point — those are two equal lengths. Give the reason each time ('tangent is perpendicular to the radius' or 'the two tangents from a point are equal'). In the kite, use ∠AOB + ∠APB = 180°, or the right-angled triangle OAP.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Use the two tangent facts

PA and PB are tangents from P, touching at A and B. The angle between them is ∠APB = 48°. Find ∠AOB, and state what you know about PA and PB.

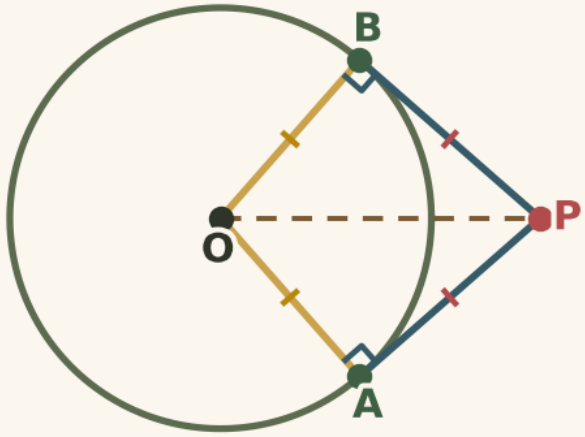


1. Tangent meets radius at 90°: ∠OAP = ∠OBP = 90°.	right angles at A, B
2. Kite OAPB angles sum to 360°, so ∠AOB + ∠APB = 180°.	∠AOB = 180 – 48
3. The two tangents from a point are equal.	PA = PB

∠AOB = 132°, and PA = PB (the two tangents from P are equal).

Example 2 — Why the two tangents are equal

Prove PA = PB for the two tangents from an external point P.



1. Tangent ⊥ radius, so ∠OAP = ∠OBP = 90°.	right angles
2. OA = OB (radii) and OP is shared.	equal radii, shared side
3. Right angle, hypotenuse OP, one equal side ⇒ congruent (RHS).	triangles OAP ≅ OBP

So PA = PB — the two tangents from an external point are equal. ■

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (what a tangent is (Lesson 1), that all radii of a circle are equal, congruent triangles (RHS), and that angles in a triangle sum to 180°), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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